

With these blocks, you need to attach metal plates to each of the RAM modules and then place the block atop the modules which are then screwed into the RAM block. The metal plates on the RAM conduct heat towards the RAM block which in turn is liquid cooled.

Motherboard VRM

The CPU attached to your motherboard doesn't directly draw power from the PSU. It's passed through the VRM (Voltage Regulator Module) circuitry which is made up of several phases. Each phase remains operational for a fraction of a second before switching itself off and turning the next phase on. This way the workload is spread across multiple phases and the VRM circuitry doesn't go up in smoke. The VRM is one of the hottest components apart from the CPU and the motherboard chipset. This is why you'll find heatsinks on the VRM circuitry on most mid-range and higher motherboards. Some like the ASUS MAXIMUS X Formula even have built-in water cooling capabilities on the VRM heatsink.

When you overclock the CPU, you end up drawing more power through the VRM circuitry and as a result they heat up even further.

That's when water cooling helps. At higher temperatures, the MOSFETs used for the VRM don't operate as expected because the heat alters the physical properties. Consequently, the VRM tends to falter and you don't get a stable supply of power to the CPU. At high overclocks, a stable power supply to the CPU is extremely important. So you can see why water cooling is more of an enthusiast need rather than just a way to fancy up your motherboard.



ASUS MAXIMUS X Formula



The anatomy of a VRM block